Decent Coaching Classes

VIII CBSE

Mind Your Math - IX

ANSWER KEY & SOLUTIONS

Section A — MCQ (1 mark each) **[6]**

(1) A (2) C (3) D (4) A (5) B (6) C

Hints: (1) 0.35 × 240 = 84. (2) 0.045 × 100 = 4.5%. (3) 1.2 × 0.8 = 0.96 of the original, so a 4% decrease. (4) Discount = 120, SP = 800 − 120 = ₹ 680.

(5) Loss = 50; loss % = 50/500 × 100 = 10%. (6) SP = 640 × 125/100 = ₹ 800.

Section B — (2 marks each) **[8]**

7. 7/20 × 100 = 7 × 5 = **35%**
8. Required percent = 36/150 × 100 = **24%**
9. Increase = 8800 − 8000 = 800. Percentage increase = 800/8000 × 100 = **10%** [1 + 1]
10. Loss = 1200 − 1050 = **₹ 150**. Loss % = 150/1200 × 100 = **12.5%** [1 + 1]

Section C — (3 marks each) **[12]**

11. Discount = 12% of 1500 = 12/100 × 1500 = **₹ 180**. [1½] Selling price = 1500 − 180 = **₹ 1320**. [1½]
    OR: Discount = 2400 − 2040 = ₹ 360. [1] Rate of discount = 360/2400 × 100 [1] = **15%** [1]
12. Cost price = 20,000 + 1,000 = ₹ 21,000. [1] Profit = 20% of 21,000 = ₹ 4,200. [1] Selling price = 21,000 + 4,200 = **₹ 25,200**. [1]
    OR: SP = ₹ 1680 for a 12% gain, so 112% of CP = 1680. [1] CP = 1680 × 100/112 [1] = **₹ 1500** [1]
13. After 10% increase: 1200 × 110/100 = ₹ 1320. [1] After 10% decrease: 1320 × 90/100 = ₹ 1188. [1] Final price = **₹ 1188**; it is not the same as the original; it is ₹ 12 less (a 1% net decrease). [1]
    OR: Mathematics: 72/90 × 100 = 80%. [1] Science: 68/80 × 100 = 85%. [1] He performed better in **Science** (85% > 80%). [1]
14. Let CP = ₹ 100. Marked price = 100 + 40 = ₹ 140. [1] Discount = 10% of 140 = ₹ 14, so SP = 140 − 14 = ₹ 126. [1] Profit = 126 − 100 = ₹ 26, so profit = **26%**. [1]
    OR: GST = 18% of 500 = 18/100 × 500 = **₹ 90**. [2] Total amount = 500 + 90 = **₹ 590**. [1]

Section D — Case-Based (4 marks) **[4]**

15. (i) First discount = 20% of 2500 = ₹ 500. Price = 2500 − 500 = **₹ 2000**. [1]
    (ii) Additional discount = 10% of 2000 = ₹ 200. Anita pays 2000 − 200 = **₹ 1800**. [1]
    (iii) Total discount = 2500 − 1800 = **₹ 700**. [1] Single discount % = 700/2500 × 100 = **28%**. It is not equal to 30%, because the second 10% is taken on the reduced price (₹ 2000), not on ₹ 2500. [1]

Note: Award full marks for any correct alternative method.